Amrik Verma

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Education

University of California, Berkeley GPA: 3.70
B.S. Mechanical Engineering, Minor in EECS *May 2028*

Coursework: Control Systems, Dynamics, Solid Mechanics, CAD, Python & MATLAB, Circuits, Thermodynamics, Experimentation & Measurements, Fluid Mechanics, Mechanics of Materials, Manufacturing, Linear Algebra, Diff Eqs, DSA

Experience

Lucid Motors Newark, CA
CAE Engineering Intern *May 2026 – Aug 2026*

- Built and validated low content LS-DYNA crash models in ANSA and META for accurate, faster multi-impact crash prediction
- Optimized structural crash components using FEA, achieving up to 50% weight reduction while preserving crashworthiness
- Ran nonlinear crash simulations to identify failure modes and optimize structural crashworthiness for passenger safety
- Correlated CAE results with physical crash tests to improve model fidelity and accelerate simulation-driven design decisions

FSAE Berkeley Formula Racing Berkeley, CA
Drivetrain Engineer *Sept 2025 – Present*

- Optimized 7075-T6 aluminum wheel centers for carbon fiber shells with SOLIDWORKS and ANSYS topology optimization to achieve a 59% weight reduction, validating it to a factor of safety of 1.3 with FEA
- Simulated the wheel hub stackup assemblies with ANSYS Static Structural analysis under loads exceeding 3.5 kN, reducing the weight by 20% in unsprung mass and validating a transition from a 1-cylinder to a 4-cylinder powertrain
- Diagnosed chainguard resonance via ANSYS modal analysis, raising natural frequency 82% and fatigue FOS from 1.9 to 3.5
- Designed and CNC-machined custom AISI 4130 hex standoff bolts, eliminating undamped engine vibration at a 4.9% mass cost

University of California, Berkeley, Department of Mechanical Engineering Berkeley, CA
Undergraduate Researcher - High Performance Robotics Lab & FLOW Lab *May 2025 – Present*

- Implemented NMPC for tiltable-quadcopter in PyTorch, simulating PyBullet tracking with 5% trajectory error across 50 flights
- Prototyped teaching quadcopter for ME136 UAV course, replacing Crazyflie drones with safer, efficient, and 15% cheaper design
- Collaborated with Professor Simo Mäkiharju to design, optimize, and construct a Vertical Air-Water Flow Loop with 24 sensors

Ati Motors Detroit, MI
Mechanical Engineering Intern *May 2024 – Aug 2024*

- Utilized PLM software to manage all aspects within the product lifecycle of various battery modules, industrial sensors, and actuators including design, development, manufacturing, and validation
- Assisted with engineering change orders (ECOs) by modifying drawing files for bumpers and wheel casters whilst adhering to GD&T standards (ASME Y14.5)

Projects

Voice-Controlled Robotic Hand | *Robotics, IoT, Circuitry, Onshape, 3D Printing* *Jan 2026 – May 2026*

- Engineered a 5-finger servo-driven robotic hand on ESP32 with offline voice recognition, executing 6 biomechanical gestures
- Architected an IoT pipeline via MQTT and REST API, streaming live finger telemetry and hand state to a remote dashboard
- Designed a dual-rail NiMH power system with braided tendon actuation and M2 steel joints for precise, repeatable finger articulation in a PLA chassis

Wind Turbine Design Project | *SOLIDWORKS, 3D Printing, FEA* *Jan 2025 – May 2025*

- Developed a 3D-printed wind turbine generating 2W, optimized stiffness and assessed stress distribution through linear FEA
- Designed rotor blades with 10 degrees angle of attack, 17 degrees twist, and 17 N/mm stiffness, achieving 29% energy increase

Leadership

Pi Tau Sigma - UC Berkeley's Mechanical Engineering Honor Society Berkeley, CA
President *May 2025 – Present*

- Established relationships with sponsors including Boeing and General Motors, securing $10,000 in funding and resources, significantly enhancing club growth and member opportunities
- Led and coordinated 20+ professional, research, and social events annually for members increasing student participation by 40%

Skills

Technical: FEA, Topology Optimization, ROS, Circuitry, Arduino, CAD, GD&T, 3D Printing, Machining, R&D, DFM, Git
Programs: SOLIDWORKS, ANSYS, Simulink, LS-DYNA, ANSA, META, LabVIEW, CATIA 3DX, Fusion 360, OnShape, PDM
Languages: Python, MATLAB, C++, Java, PyTorch, Pandas, NumPy, LaTeX